

Lecture notes week 2

Semi-discrete formulation of elastodynamics.

In linear elastodynamics, the displacement field $u(\mathbf{x}, t)$ satisfies the strong form:

$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}},$$

where $\mathbf{C}$ is the elasticity tensor, $\mathbf{f}$ the body force, and ρ the density.

We obtain the weak form by multiplying by a virtual displacement $\delta \mathbf{u}$ and integrating over the domain:

$$\int_{\Omega} (\nabla \delta \mathbf{u})^T \mathbf{C} \nabla \mathbf{u} d\Omega + \int_{\Omega} \rho \delta \mathbf{u}^T \ddot{\mathbf{u}} d\Omega = \int_{\Gamma} \delta \mathbf{u}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \delta \mathbf{u}^T \mathbf{f} d\Omega.$$

Using the Galerkin method, we approximate:

- $\mathbf{u}(\mathbf{x}, t) \approx \mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x}) \mathbf{q}(t)$,
- $\delta \mathbf{u}(\mathbf{x}) \approx \delta \mathbf{u}^h(\mathbf{x}) = \mathbf{H}(\mathbf{x}) \delta \mathbf{q}$,

where $\mathbf{H}(\mathbf{x})$ contains the shape functions and $\mathbf{q}(t)$ the nodal degrees of freedom.

This yields the semi-discrete weak form:

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{r}(t),$$

with:

$$\begin{aligned} \mathbf{M} &= \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega, \\ \mathbf{K} &= \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega, \\ \mathbf{r}(t) &= \int_{\Gamma} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega. \end{aligned}$$

Here, $\mathbf{B} = \nabla \mathbf{H}$ is the strain-displacement matrix.

Initial conditions

Given $\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x})$, $\dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{v}_0(\mathbf{x})$, the initial values for $\mathbf{q}(t)$ and $\dot{\mathbf{q}}(t)$ are:

$$\mathbf{q}(0) = \mathbf{u}_0(\mathbf{x}_j), \quad \dot{\mathbf{q}}(0) = \mathbf{v}_0(\mathbf{x}_j).$$

Boundary conditions

On Γ_u , we enforce:

$$\mathbf{q}_j(t) = \hat{\mathbf{u}}(\mathbf{x}_j, t), \quad \delta \mathbf{q}_j = 0.$$

Advantages

- Systematic approximation method.
- Reuses the same shape functions for test and trial functions.
- Converges efficiently with suitable shape functions.

Drawbacks

- Heavily dependent on the chosen basis.
- No direct physical interpretation of $q(t)$.
- Boundary and initial conditions may be cumbersome in practice.